

Philip Loewen

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Education

University of Toronto, Toronto, ON, Canada 2026
Master of Science, Statistics | 3.9/4.0 GPA

University of British Columbia, Vancouver, BC, Canada 2025
Bachelor of Science, Mathematics | 85.8/100 GPA | Dean's List | Stanley M. Grant Scholarship in Mathematics

Research

Graduate Researcher, University of Toronto | *Supervisor: Dr. Joshua Speagle* Sep. 2025 - Present

- Designed and executed a large-scale benchmarking study evaluating six Time Series Foundation Models across 400K+ samples spanning finance, healthcare, audio, and astronomy. Engineered domain-specific feature extractors achieving 90.3% accuracy against a 67.9% foundation model ceiling.
- Built end-to-end validation pipelines in Python and PyTorch using embedding extraction, PCA-based dimensionality reduction, and a multi-head classification suite. Identified geometric collapse as the primary failure mode in high entropy signal domains.
- Studied theory of deep learning models through Neural Tangent Kernels, Neural Network Gaussian Process and muP.

Experience

Statistical Consultant - Ontario Ministry of Natural Resources Nov. 2025 - Present

University of Toronto, Department of Statistical Sciences, Toronto, Canada

- Partnered with the Ontario Ministry of Natural Resources to develop a hierarchical zooplankton classification system.
- Conducted stakeholder meetings to define requirements and align technical approach with team goals.
- Implemented and trained a hybrid CNN-RNN hierarchical image classifier on 280K samples via sequence prediction over taxonomic tree paths in Python and PyTorch, achieving 95% leaf and 99% parent level accuracy.

Teaching Assistant - Probability, Statistics, and Data Analysis I, II

Sep. 2025 - Present

University of Toronto, Department of Statistical Sciences, Toronto, Canada

- Delivered tutorial instruction to 60 students per session across two quantitative courses.
- Provided written feedback on assignments and exams for a cohort of 1200 students.

Landscaper, Paraspace Landscaping, Vancouver, Canada

May - Aug. 2022, 2023

Landscaper working in team environment, to complete high-volume tasks with rotating site supervisors.

Projects

Fintech Ledger Simulator | *Python, PostgreSQL*

- Designed and queried a multi-table PostgreSQL schema using CTEs, window functions, and self-joins to analyze transaction networks, geographic wealth distribution, and cyclic arbitrage opportunities.
- Seeded database with Python using log-normal distributions to simulate realistic wealth inequality.

nuScenes Trajectory Prediction | *Python, PyTorch*

- Implemented and compared RNN and Transformer architectures for 2D vehicle and pedestrian trajectory prediction.
- Developed a standardized preprocessing pipeline for normalizing positions, handling missing data, and segmenting sequences for fair cross-model comparison.

Internet Speed Experiment | *Experimental Design, R, ANOVA*

- Designed and conducted a full factorial ANOVA experiment (81 observations) to evaluate campus WiFi reliability.
- Identified optimal upload conditions with 90% confidence, projecting peak speeds of 221.9 Mbps.

Relevant Skills

Languages: English (Native), French (Native), Spanish (Basic)

Software: Python (PyTorch, Pandas, SciPy, Scikit-Learn Matplotlib), R, SQL, PostgreSQL, Git, JAX, Spark

Statistics: Regression, Classification, Clustering, Dimension Reduction, Time Series Analysis, Bayesian Methods (MCMC, Variational Inference, Gaussian Processes), Stochastic Processes, Experimental Design, Causal Inference, Nonparametric Inference, Optimization (Convex, Linear)

Machine Learning: Deep Learning, Foundation Models, XGBoost, Decision Trees, Feature Engineering, Benchmarking, Ensemble Methods, Model Versioning, CI/CD